

Rising Sea Week 2 Tutorial

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Part I. Adjoints

Exercise 1 (Tensor–Hom from the universal property). *Let A be a commutative ring and let M, N, P be A -modules.*

- (a) *Given $\Phi : M \otimes_A N \rightarrow P$, define a map $\Theta(\Phi) : M \rightarrow \text{Hom}_A(N, P)$ by specifying $\Theta(\Phi)(m)(n)$. Check that $\Theta(\Phi)$ is A -linear.*
- (b) *Conversely, given $f : M \rightarrow \text{Hom}_A(N, P)$, construct an A -linear map $\Psi(f) : M \otimes_A N \rightarrow P$. Your construction should use the universal property of the tensor product.*
- (c) *Show that Θ and Ψ are inverse bijections. Hence obtain*

$$\text{Hom}_A(M \otimes_A N, P) \cong \text{Hom}_A(M, \text{Hom}_A(N, P)).$$

- (d) *Fix N . Identify explicitly the unit and counit of the tensor-hom adjunction.*

Exercise 2 (Groupification as a left adjoint). *Let S be a nonempty abelian semigroup, written additively. On $S \times S$ define*

$$(a, b) \sim (c, d) \iff \text{there exists } e \in S \text{ such that } a + d + e = b + c + e.$$

- (a) *Show that $\sim$ is an equivalence relation.*
- (b) *Let $H(S) := (S \times S)/\sim$, and write $[a, b]$ for the class of (a, b) . Show that $[a, b] + [c, d] := [a + c, b + d]$ is well-defined and makes $H(S)$ into an abelian group.*
- (c) *Show that*
$$\eta_S : S \rightarrow H(S), \quad a \mapsto [a + s_0, s_0]$$
is independent of the chosen $s_0 \in S$ and is a semigroup morphism.
- (d) *Let G be an abelian group and let $f : S \rightarrow G$ be a semigroup morphism. Show that there is a unique group homomorphism $\tilde{f} : H(S) \rightarrow G$ such that $\tilde{f} \circ \eta_S = f$.*

- (e) *Deduce that groupification is left-adjoint to the forgetful functor $U : \mathbf{Ab} \rightarrow \mathbf{AbSemiGrp}$.*

- (f) *Identify the groupifications of*

$$(\mathbb{Z}_{\geq 0}, +), \quad (\mathbb{Z}_{> 0}, +), \quad (\mathbb{Z}_{> 0}, \times).$$

Part II. Sheaves

Exercise 3 (Constant presheaf versus constant sheaf). Let $X = \{p, q\}$ with the discrete topology, and let S be a set with at least two elements.

Define the constant presheaf S^{pre} by $S^{\text{pre}}(U) = S$ for every nonempty open U , with identity restriction maps.

- (a) Show that S^{pre} is a presheaf.
- (b) Choose distinct $a, b \in S$. Put the section a on $\{p\}$ and the section b on $\{q\}$. Why are these sections compatible on overlaps?
- (c) Show that they cannot be glued in $S^{\text{pre}}(X)$. Hence the constant presheaf is not, in general, a sheaf.
- (d) Now define

$$\underline{S}(U) := \{f : U \rightarrow S : f \text{ is locally constant}\}.$$

Show that $\underline{S}$ is a sheaf.

- (e) Compute $\underline{S}(X)$ and the stalks $\underline{S}_p, \underline{S}_q$.
- (f) Give S the discrete topology. Show that the espace étalé of $\underline{S}$ is naturally $X \times S \rightarrow X$.

Exercise 4 (Sections of a map form a sheaf). Let $\mu : Y \rightarrow X$ be a continuous map. For each open $U \subseteq X$, define

$$\mathcal{S}_\mu(U) = \{s : U \rightarrow Y : s \text{ is continuous and } \mu \circ s = \text{id}_U\}.$$

- (a) Define the restriction maps and verify that $\mathcal{S}_\mu$ is a presheaf.
- (b) Prove the identity and gluability axioms. Hence $\mathcal{S}_\mu$ is a sheaf.
- (c) Suppose now that μ is a local homeomorphism. Fix $x \in X$. Show that every $y \in \mu^{-1}(x)$ determines a germ of a local section at x .
- (d) Show that $(\mathcal{S}_\mu)_x \cong \mu^{-1}(x)$.
- (e) Conclude that, when μ is a local homeomorphism, the espace étalé of $\mathcal{S}_\mu$ recovers $Y \rightarrow X$ up to the evident canonical identification.

Exercise 5 (The sheaf condition as a limit). Let $\mathcal{F}$ be a presheaf of sets on X , and let $U = \bigcup_{i \in I} U_i$ be an open cover. Define

$$\alpha, \beta : \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

by

$$\alpha((s_i)_i)_{ij} = s_i|_{U_i \cap U_j}, \quad \beta((s_i)_i)_{ij} = s_j|_{U_i \cap U_j}.$$

- (a) Show that restriction gives a natural map

$$\rho : \mathcal{F}(U) \rightarrow \prod_i \mathcal{F}(U_i).$$

- (b) Show that the identity axiom is equivalent to ρ being injective.
- (c) Show that the gluability axiom is equivalent to the image of ρ being precisely

$$\{(s_i)_i : \alpha((s_i)) = \beta((s_i))\}.$$

(d) Conclude that $\mathcal{F}$ is a sheaf exactly when, for every open cover,

$$\mathcal{F}(U) \cong \text{Eq} \left(\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j) \right).$$

(sorry! You may need to look up what an equaliser is)

Exercise 6 (Stalks of a skyscraper sheaf). Fix $p \in X$ and a set S . Define

$$\mathcal{K}(U) = \begin{cases} S, & p \in U, \\ \{*\}, & p \notin U. \end{cases}$$

- (a) Write down all restriction maps and verify directly that $\mathcal{K}$ is a sheaf.
- (b) Suppose X is T_1 (in particular, Hausdorff). Compute $\mathcal{K}_x$ for every $x \in X$.
- (c) More generally, show that $\mathcal{K}_x \cong S$ exactly when every neighbourhood of x contains p .
- (d) Let $X = \{0, 1\}$ with topology $\{\emptyset, \{1\}, X\}$, and take $p = 1$. Compute both stalks. What does this say about the informal phrase “all the information is concentrated at p ”?

Exercise 7 ($\mathcal{O}_X$ -modules and fibres of vector bundles). Let X be a smooth manifold and let $\mathcal{O}_X(U) = C^\infty(U, \mathbb{R})$. Let $\pi : E \rightarrow X$ be a smooth vector bundle of rank r , and let $\mathcal{E}$ denote its sheaf of smooth sections.

- (a) Verify that $\mathcal{E}$ is naturally an $\mathcal{O}_X$ -module.
- (b) If $E|_U \cong U \times \mathbb{R}^r$ is a trivialisation, show that $\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus r}$ as $\mathcal{O}_U$ -modules.
- (c) Fix $p \in X$. Deduce that $\mathcal{E}_p \cong \mathcal{O}_{X,p}^{\oplus r}$.
- (d) Let $\mathfrak{m}_p \subset \mathcal{O}_{X,p}$ be the maximal ideal of germs vanishing at p . Show that evaluation at p induces a natural vector-space isomorphism $\mathcal{E}_p / \mathfrak{m}_p \mathcal{E}_p \cong E_p$.

I will share some solutions at some point, hopefully!